

# Revision HQ 2.2

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | OCR J277 2.2 Programming fundamentals — Paper 2, Computational thinking, algorithms and programming               |
| Lesson          | Revision HQ 2.2 (lesson 7 of the unit)                                                                            |
| Lesson ID       | pf-l07                                                                                                            |
| Experience ID   | pf-l07                                                                                                            |
| Plan ID         | PLAN-PF-L07                                                                                                       |
| Content version | 9e994b1a0c60                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=pf-l07">https://revise360.co.uk/experience.html?id=pf-l07</a> |

## Big question, outcomes and success criteria

**Big question** — Which parts of 2.2 am I secure on, and which do I need to revise before the test?

### Prerequisites

The lesson before this one is pf-l06. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: pf-l01, pf-l02, pf-l03, pf-l04, pf-l05.

### Assessable outcomes

1. **pf-l07-o1** — know the whole of topic 2.2 well enough to answer without notes
2. **pf-l07-o2** — understand the mistakes that most often lose marks in programming questions
3. **pf-l07-o3** — know which lessons are secure and which still need revision before the test

## Vocabulary, resources and preparation

**Key vocabulary** — Iteration, Pseudo-random

### Resources

|                   |                                               |
|-------------------|-----------------------------------------------|
| SLIDES            | presentations/PF_L07_RevisionHQ22_Lesson.pptx |
| WORKSHEET         | worksheets/PF_L07_RevisionHQ_Worksheet.docx   |
| ONLINE EXPERIENCE | experience.html?id=pf-l07                     |

### Before the lesson

1. Sign in on the **teacher dashboard** (`teacher.html`) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

### How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                              | TEACHER                                                                                      | PUPILS                                                                          | EVIDENCE TO CHECK                                                                      |
|-------|------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0-5   | Retrieve                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                                            | The starter box on PF_L07_RevisionHQ_Worksheet.docx is filled in.                      |
| 5-8   | Review the starter                 | Take answers; correct the one most got wrong.                                                | Correct their own starter.                                                      | A correction written, not just a tick.                                                 |
| 8-18  | Explain and model                  | The new idea in two or three steps, with the example worked in front of the class.           | Listen and answer questions about the example.                                  | They can say why, not only what.                                                       |
| 18-21 | Set up and settle                  | Sign-in, launch, and what to write before anyone taps an answer.                             | Sign in, open the lesson, find Constructs recap.                                | Everyone is in and on station 1.                                                       |
| 21-43 | Online: work the stations in order | Pupils read each wall, write the key fact and the challenge, then answer at the badge.       | Read each wall, write the key fact and the challenge, then answer at the badge. | The worksheet has writing on it for each of the 6 stations before the badge is tapped. |
| 43-48 | Check understanding                | The hinge question, hands down.                                                              | Answer the hinge question, hands down.                                          | The split across the options.                                                          |
| 48-56 | Apply                              | Exam practice on the worksheet, in silence.                                                  | Exam practice, in silence.                                                      | worksheet, Exam practice a) to c)                                                      |
| 56-60 | Exit and set up the next lesson    | The exit question and the retrieval link to next lesson.                                     | Write the exit answer.                                                          | One sentence that answers the big question.                                            |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. Constructs recap

- A single equals sign assigns a value to a variable; a double equals sign compares two values and gives True or False.
- Selection makes a choice once. Iteration goes back and repeats.
- A condition-controlled loop must change the variable in its condition somewhere inside the loop, or it never ends.

**Challenge on this wall:** Write down the difference between `=` and `==` and give an example of each.

### 2. Types and operators recap

- In the OCR Exam Reference Language, `+` joins two strings but adds two numbers, so `"7" + "2"` gives the string `72` while `7 + 2` gives `9`.
- `input()` always returns a string, so a number must be cast before any arithmetic is done with it.
- `DIV` gives how many whole times one number goes into another, and `MOD` gives the remainder left over.

**Challenge on this wall:** Explain what `int(7.9)` gives, and why that is not the same as rounding.

### 3. File handling recap

- In Python 3, mode `"w"` wipes whatever was in the file, and mode `"a"` adds to the end and keeps what was already there.
- In the OCR Exam Reference Language a read loop calls `readLine()` inside the loop and stops when `endOfFile()` becomes True.
- A file must be closed, or the last data written may never leave the buffer in memory.

**Challenge on this wall:** A program saves a new high score each week, but the file only ever holds one score. Explain what is wrong and how to fix it.

#### 4. Records and SQL recap

- SELECT chooses the fields, FROM chooses the table and WHERE chooses the records.
- Text in a WHERE condition goes inside single quotes, and a pattern with a wildcard needs LIKE rather than =.
- AND needs both of its conditions to be True, so it returns the same number of records as OR or fewer.

**Challenge on this wall:** Write down the three parts of an SQL query and say what each one decides.

#### 5. Arrays and sub-programs recap

- An array of n elements is indexed 0 to n - 1, so the last index is always one less than the size.
- A function returns a value to the line that called it, and a procedure does not.
- A local variable only exists inside its own sub-program, so no other part of the program can read it.

**Challenge on this wall:** An array holds twelve lane numbers. Give the index of the first and of the last, and write the loop that would visit them all.

#### 6. Randomness recap

- In the OCR Exam Reference Language, random(1, 6) includes both 1 and 6.
- A computer produces pseudo-random numbers with an algorithm and a seed, so the same seed gives the same sequence.
- A random index must stay inside the array, so an array of six elements needs a number from 0 to 5.

**Challenge on this wall:** Explain why random(0, 6) is the wrong call for choosing one of six prizes from an array.

### Retrieval and the check for understanding

**Starter** — Without looking back at your notes, write down the three programming constructs, the five data types, and the four operations a program carries out on a file.

Answer space: 4 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

#### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pf - 107 - s4 - c1), so the wording a pupil sees matches the wording you say.

**A table called Pupils has the fields Name, Year and Choir. Which query returns the names of pupils who are in Year 10 and in the choir?**

**Correct:** SELECT Name FROM Pupils WHERE Year = 10 AND Choir = True;

#### What a pupil might choose instead

- SELECT Name FROM Pupils WHERE Year = 10 OR Choir = True;
- SELECT Year, Choir FROM Pupils WHERE Name = 'Year 10';
- SELECT Name FROM Pupils LIKE Year = 10 AND Choir = True;

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** Both conditions must be True at the same time, and AND is the operator that demands that.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

### Guided to independent practice

| STAGE     | ACTIVITIES | WHAT IT IS                                                                                                                |
|-----------|------------|---------------------------------------------------------------------------------------------------------------------------|
| Recognise | 14         | The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.                   |
| Construct | 1          | Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer. |

## Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

## Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| #     | STATION                        | ACTIVITIES | RECORD ON PAPER                                                |
|-------|--------------------------------|------------|----------------------------------------------------------------|
| 1     | Constructs recap               | 2          | A single equals sign... A double equals sign...                |
| 2     | Types and operators recap      | 2          | Casting is needed because...                                   |
| 3     | File handling recap            | 2          | The mode a file is opened in decides...                        |
| 4     | Records and SQL recap          | 2          | The part of a query that decides which records come back is... |
| 5     | Arrays and sub-programs recap  | 2          | An index goes out of range when...                             |
| 6     | Randomness recap               | 2          | Pseudo-random numbers come from...                             |
| final | Final challenge: the lane draw | 3          | the answer to the station's challenge                          |

### Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

### Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

## Assessment, evidence and unfinished work

| ID                | QUESTION                                                                                                                                                                                                                        | MARKS | WHERE                                                        |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------|
| pf-<br>l07-<br>x1 | State the difference between a count-controlled loop and a condition-controlled loop.                                                                                                                                           | 2     | worksheet, Exam practice<br>a); deck, Exam practice slide    |
| pf-<br>l07-<br>x2 | A club stores its members in a table called Members, with the fields MemberID, Name, Year and Paid. Write an SQL query that returns the name of every member in Year 10 who has not paid.                                       | 3     | worksheet, Exam practice<br>b); deck, Exam practice<br>slide |
| pf-<br>l07-<br>x3 | Eight runners' names are stored in a file. Write an algorithm that gives each runner a different lane from 1 to 8, chosen at random, and outputs the finished draw. Use the OCR Exam Reference Language or a language you know. | 6     | worksheet, Exam practice<br>c); deck, Exam practice slide    |

### Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/pf-l07.json`, which is not published with the website.

### Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Which of these lines compares two values rather than storing one?** — `total = 20`; `total = total + 20`
- **A loop is written as `while lives > 0` but nothing inside it ever chan** — The loop runs exactly once; The loop is skipped altogether
- **In the OCR Exam Reference Language, what is the value of `"7" + "2"` ?** — The integer 9; The real number 7.2
- **What does `int(7.9)` give?** — 8; 7.9
- **A program outputs the first line of a file over and over instead of wo** — The file was opened in append mode; The file was never closed

### Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

## Exit, next lesson and homework

### Exit check

One of these, written, not a show of hands:

- Variables, constants and the three constructs
- Data types, casting, operators and string handling
- Files, records and SQL, arrays, sub-programs and random numbers

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### Retrieval link to next lesson

Next is `pf-l08-test`, End-of-topic test. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

### Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## Shorter lessons, longer lessons and no internet

### A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

### If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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